Sungwon Yoon

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PROFESSIONAL SUMMARY

- Ph.D. experimental chemist in organometallic, inorganic, and materials chemistry with recent, sustained hands-on lab work, from multistep synthesis and purification through full characterization.
- Hypothesis-driven researcher who designs compound series and uses kinetic and spectroscopic studies to explain, predict, and optimize molecular properties, leading to a first-author *JACS* paper and deep-red phosphors with quantum yields up to ∼0.5.
- Adaptable across compounds, materials, and formulations, with work spanning organometallic complexes, nanomaterials, thin films, and battery electrode slurries that relates composition and structure to properties.
- Industry R&D background with a focus on high-quality, well-documented data, troubleshooting, and process transfer.
- Independent and cross-functional, leading projects from experimental design through publication across a university and two national laboratories with 13 peer-reviewed papers (6 first-author).

SKILLS

Synthesis & Purification: Ir(III)/Pt(II)/Cu(I) organometallic and coordination chemistry, ligand design, air-free synthesis (Schlenk line, glovebox), column chromatography, recrystallization, ZnO quantum dot and MOF synthesis, surface functionalization
Kinetics & Spectroscopy: steady-state and time-resolved (TCSPC) photoluminescence, photoluminescence quantum yields, excited-state decay kinetics, UV-vis (calibration-curve quantitation), transient absorption (collaborative)
Characterization: NMR, ESI-MS, FTIR, PXRD, SEM, cyclic voltammetry (solvent/electrolyte selection, ferrocene referencing)
Data & Tools: design of experiments, structure–property analysis, analytical method development, troubleshooting and root cause analysis, documentation for reproducibility and process transfer, OriginPro, Excel, ChemDraw, SciFinder
Materials & Safety: Li-ion slurry formulation, Langmuir–Blodgett films, air-sensitive and hazardous chemical handling

EXPERIENCE

Postdoctoral Appointee 2025–Present
Argonne National Laboratory, PI: Karen L. Mulfort *Lemont, IL*

- Design and synthesize heteroleptic Cu(I)HETPHEN complexes under inert atmosphere, and measure redox potentials by cyclic voltammetry to test the thermodynamic driving force for charge transfer to ZnO.
- Synthesize ZnO quantum dots and functionalize them with Cu(I) chromophores to probe directional interfacial charge transfer.
- Interpret spectroscopic data with collaborators (including transient absorption) and communicate results in technical reports and presentations.

Graduate Research Assistant 2020–2024
University of Houston, PI: Thomas S. Teets *Houston, TX*

- Executed multistep air-free syntheses (Schlenk line, glovebox) of cyclometalated Ir(III) and Pt(II) complexes for deep-red to near-infrared phosphorescence, an emitter class used in OLEDs and explored for imaging and sensing.
- Designed systematic ligand-variation series and measured each compound's redox potentials, emission, quantum yield, and excited-state lifetime to establish quantitative structure–property relationships for excited-state decay.
- Wrote a first-author *Chem. Commun.* Feature Article reviewing the group's design rules linking molecular structure, frontier orbital energies, and excited-state dynamics.

R&D Chemist 2018–2019
Nanotech Energy *Chico, CA*

- Developed and optimized Li-ion electrode formulations and slurry processing, including tuning slurry viscosity, to improve coating quality, electrochemical performance, and cycling efficiency.
- Coated electrodes, built and tested lab-scale cells, and documented results for reproducibility and process transfer.

Undergraduate Research Assistant 2017–2018
California State University, Chico, PI: Monica C. So *Chico, CA*

- Synthesized metal–organic frameworks (MOFs) for dye removal from water, evaluated aqueous adsorption and regeneration, and characterized the materials by PXRD, SEM, FTIR, and UV-vis.

Undergraduate Research Intern Summer 2017, 2018
Sandia National Laboratories *Livermore, CA*

- Fabricated metal–organic monolayer films by Langmuir–Blodgett deposition and synthesized Ni- and Pt-based conductive MOFs for a multi-institution study showing that metal substitution switches charge transport from n- to p-type.

SELECTED PUBLICATIONS

13 peer-reviewed publications (6 first-author) and a book chapter.

- **Yoon, S.**; Teets, T. S. "Ancillary Ligand Steric Effects and Cyclometalating Ligand Substituents Control Excited-State Decay Kinetics in Red-Phosphorescent Platinum Complexes." *J. Am. Chem. Soc.* **2024**, 146, 28407–28413.
- **Yoon, S.**; Teets, T. S. "Enhanced Deep Red to Near-infrared (DR-NIR) Phosphorescence in Cyclometalated Iridium(III) Complexes." *Inorg. Chem. Front.* **2022**, 9, 6544–6553.
- **Yoon, S.**; Talin, A. A.; Stavila, V.; Mroz, A. M.; Bennett, T. D.; He, Y.; Keen, D. A.; Hendon, C. H.; Allendorf, M. D.; So, M. C. "From n- to p-Type Material: Effect of Metal Ion on Charge Transport in Metal–Organic Materials." *ACS Appl. Mater. Interfaces* **2021**, 13, 52055–52062.
- **Yoon, S.**; Calvo, J.; So, M. C. "Removal of Acid Orange 7 from Aqueous Solution by Metal-Organic Frameworks." *Crystals* **2019**, 9, 17.

SELECTED PRESENTATIONS

- **Yoon, S.**; Bain, D. C.; Chen, L. X.; Mulfort, K. L. "Directional Charge Transfer in Heteroleptic Cu(I) Complexes on Metal Oxides." *ACS National Meeting*, McCormick Place Convention Center, Chicago, IL, August 2026. (Oral)
- **Yoon, S.**; Bain, D. C.; Chen, L. X.; Mulfort, K. L. "Directional Charge Transfer in Cu(I) Donor–Metal Oxide Acceptor Assemblies." *Electron Donor–Acceptor Interactions Gordon Research Conference*, Salve Regina University, Newport, RI, July 2026. (Poster)
- **Yoon, S.**; Teets, T. S. "The Effect of Ancillary Ligand on Deep-Red to Near-Infrared Emitting Bis-Cyclometalated Ir(III) Complexes." *Photochemistry Gordon Research Conference*, Bates College, Lewiston, ME, August 2023. (Poster)
- **Yoon, S.**; So, M. C. "Effects of Chloride Concentration in Graphene Analogues for Electronic Devices." *ACS National Meeting*, New Orleans Convention Center, New Orleans, LA, March 2018. (Poster)

EDUCATION

Chemistry, Ph.D. 2024
University of Houston *Houston, TX*
- Thesis: Designing Cyclometalated Iridium(III) and Platinum(II) Complexes for Efficient Red to Near-Infrared Phosphors

Chemistry, B.S. (with Honors), Minor in Mathematics 2018
California State University, Chico *Chico, CA*

AWARDS

- Jay K. Kochi Graduate Research Fellowship, University of Houston, 2024
- NSM Graduate Student Conference Travel Award, University of Houston, 2023
- CSU Program for Education and Research in Biotechnology, CSU Chico, 2017, 2018
- Student Awards for Research and Creativity (SARC), CSU Chico, 2017
- Grover Willis Award, CSU Chico, 2017